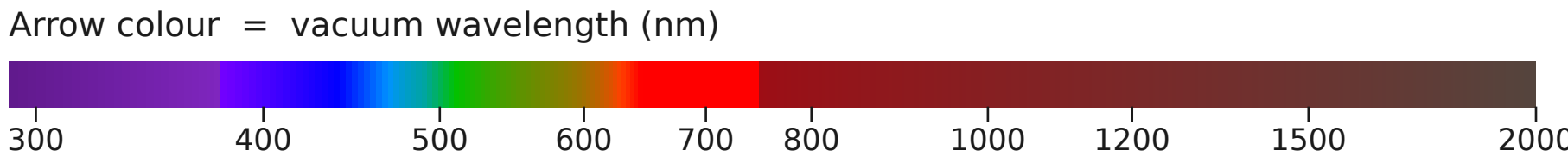
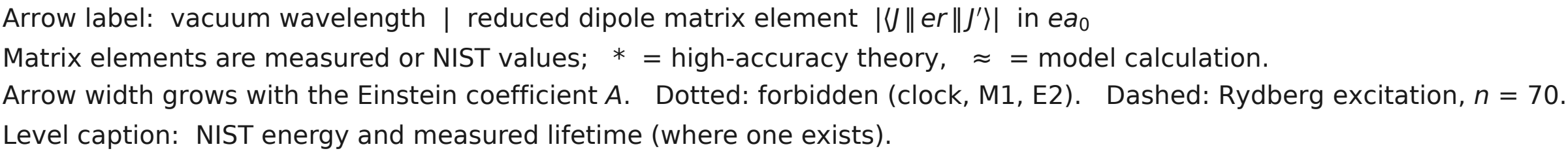


Ca⁺ ionisation limit 49305.92 cm⁻¹ = 6.11315 eV (202.815 nm)



Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J \mathbf{e} J \rangle $ (ea ₀)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
4s ² ¹ S ₀ - 4s4d ¹ D ₂	268.1088	268.0292	1118.1745	-	-	2.526	-	two-photon transition
4s ² ¹ S ₀ - 4s5p ¹ P ₀ ^o	272.2450	272.1644	1101.1861	0.090	2.700e+05	-	-	272.16 nm line
4s ² ¹ S ₀ - 4s4p ¹ P ₀ ^o	422.7917	422.6727	709.078373	4.939	2.180e+08	34.31	100	422.67 nm
4s4p ³ P ₂ ^o - 4p ² ³ P ₂	430.3738	430.2528	696.5862	5.172	1.360e+08	28.12	77	430.252 nm
4s4p ³ P ₀ ^o - 4s4d ¹ D ₂	452.7299	452.6030	662.1883	-	-	2.526	-	452.7 nm quench line used to broaden the 657 nm transition
4s ² ¹ S ₀ - 3d4s ¹ D ₂	457.6736	457.5453	655.0355	-	4.000e+01	0.0001061	6	457.5 nm electric-quadrupole line
4s4p ³ P ₂ ^o - 4s5s ³ S ₁	616.3878	616.2173	486.3698	4.067	4.770e+07	12.94	58.7	616.2 nm line of the 4s4p 3P - 4s5s 3S1 triplet
4s ² ¹ S ₀ - 4s4p ³ P ₀ ^o	657.4594	657.2779	455.986240	0.031	2.300e+03	0.0003745	97.8	657.28 nm
3d4s ¹ D ₂ - 4s5p ¹ P ₀ ^o	671.9536	671.7681	446.1506	2.322	1.200e+07	-	-	671.77 nm
4s4p ³ P ₀ ^o - 3d4s ¹ D ₂	1506.1214	1505.7099	199.0493	0.050	3.000e+02	0.0001061	83	decay of the 1D2 resonance back to the ground state via 3...
4s4p ³ P ₂ ^o - 3d4s ¹ D ₂	1530.5285	1530.1103	195.8751	0.029	9.600e+01	0.0001061	17	loss channel of the 423 nm MOT into metastable 3P2
4s4p ³ P ₀ ^o - 3d4s ³ D ₃	1978.2171	1977.6771	151.5468	-	-	-	-	1978 nm closed transition of the metastable

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A : measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Isotope shifts			
Transition	λ vac (nm)	Isotopes	Shift (MHz)
$4s^2\ ^1S_0 - 4s4d\ ^1D_2$	268.109	42-40	1268.7(3)
$4s^2\ ^1S_0 - 4s4d\ ^1D_2$	268.109	43-40	1895.5(4)
$4s^2\ ^1S_0 - 4s4d\ ^1D_2$	268.109	44-40	2445.8(4)
$4s^2\ ^1S_0 - 4s4d\ ^1D_2$	268.109	46-40	3561.0(9)
$4s^2\ ^1S_0 - 4s4d\ ^1D_2$	268.109	48-40	4582.3(8)
$4s^2\ ^1S_0 - 4s5p\ ^1P_1^o$	272.245	42-40	967(9)
$4s^2\ ^1S_0 - 4s5p\ ^1P_1^o$	272.245	43-40	1455(9)
$4s^2\ ^1S_0 - 4s5p\ ^1P_1^o$	272.245	44-40	1879(14)
$4s^2\ ^1S_0 - 4s5p\ ^1P_1^o$	272.245	46-40	2746(16)
$4s^2\ ^1S_0 - 4s5p\ ^1P_1^o$	272.245	48-40	3528(16)
$4s^2\ ^1S_0 - 4s4p\ ^1P_1^o$	422.792	42-40	393.2(3)
$4s^2\ ^1S_0 - 4s4p\ ^1P_1^o$	422.792	43-40	611.1(5)
$4s^2\ ^1S_0 - 4s4p\ ^1P_1^o$	422.792	44-40	773.79(19)
$4s^2\ ^1S_0 - 4s4p\ ^1P_1^o$	422.792	46-40	1159.5(8)
$4s^2\ ^1S_0 - 4s4p\ ^1P_1^o$	422.792	48-40	1513.0(4)
$4s4p\ ^3P_2^o - 4s5s\ ^3S_1$	616.388	43-40	-38.2(20)
$4s4p\ ^3P_2^o - 4s5s\ ^3S_1$	616.388	48-40	-116(4)
$4s^2\ ^1S_0 - 4s4p\ ^3P_1^o$	657.459	42-40	509.5(3)
$4s^2\ ^1S_0 - 4s4p\ ^3P_1^o$	657.459	43-40	782.7(2)
$4s^2\ ^1S_0 - 4s4p\ ^3P_1^o$	657.459	44-40	996.2(5)
$4s^2\ ^1S_0 - 4s4p\ ^3P_1^o$	657.459	46-40	1481.1(7)
$4s^2\ ^1S_0 - 4s4p\ ^3P_1^o$	657.459	48-40	1922.5(8)

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope}).$$